

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 July 2026
Team ID	
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture

The architecture below covers the end-to-end flow of the ServiceNow-Based Employee Laptop Request Application — from raising a request on the Employee Service Portal, through manager/IT approval and CMDB inventory validation, to fulfilment and reporting.

Use Case: An employee submits a laptop request; it is routed for manager and IT approval, checked against CMDB inventory, fulfilled by the IT Asset team, and the employee is notified at every stage.

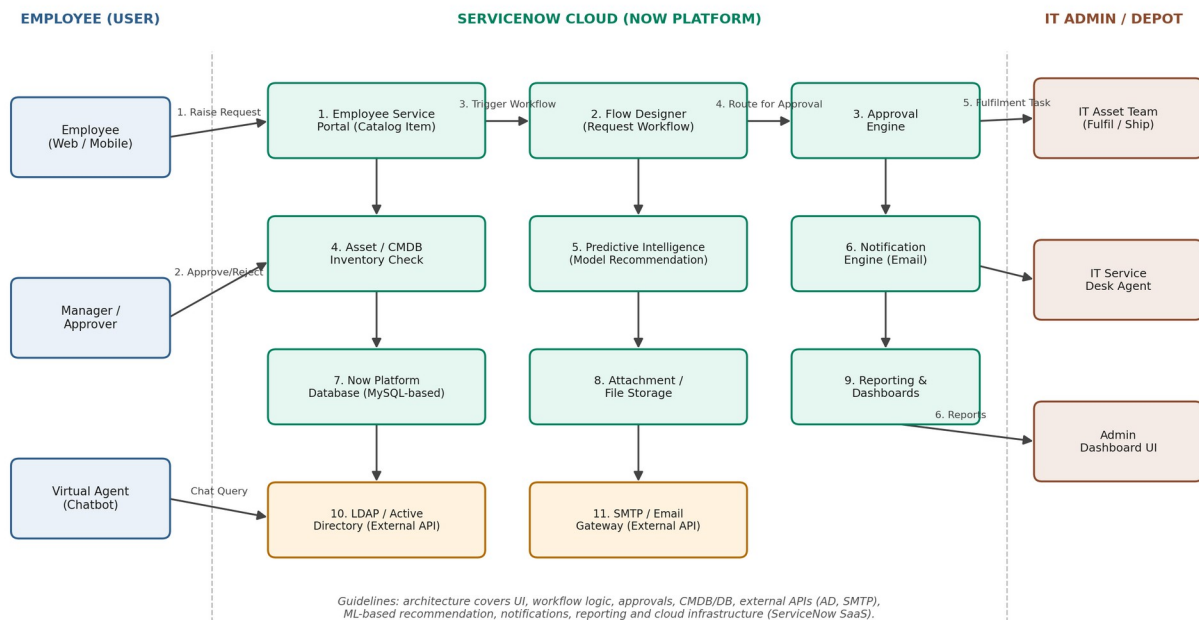


Table-1: Components & Technologies

S. No	Component	Description	Technology
1.	User Interface	How the employee interacts with the application to raise, track and manage laptop requests.	ServiceNow Service Portal (Now Experience UI), HTML, CSS, AngularJS (Portal Widgets), Mobile Agent App
2.	Application Logic-1	Catalog item intake and request-submission logic (form, variables, validations).	ServiceNow Service Catalog, Catalog Client Scripts, Glide (JavaScript)
3.	Application Logic-2	Multi-level approval logic (Manager approval, IT Head / Asset team approval).	ServiceNow Flow Designer, Workflow Editor, Approval Engine
4.	Application Logic-3	Asset availability check, auto-assignment and fulfilment task logic.	ServiceNow Asset Management, CMDB, Fulfiller Task Automation
5.	Database	Data type, configuration and storage of request records, approvals and asset data.	ServiceNow internal MySQL-based data store, GlideRecord tables
6.	Cloud Database	Database service hosted on cloud for the Now Platform instance.	ServiceNow Now Platform Cloud (Multi-Instance Architecture)
7.	File Storage	Storage for attachments such as ID proof, delivery acknowledgement, invoices.	ServiceNow Attachment Storage, Azure Blob Storage
8.	External API-1	Validating employee identity, department and reporting-manager details.	LDAP / Active Directory Integration API, SAML SSO
9.	External API-2	Sending automated notifications and status updates to employees and approvers.	SMTP / Microsoft Office 365 Email API, MS Teams Notification API
10.	Machine Learning Model	Recommending suitable laptop configuration by role/department and predicting approval likelihood.	ServiceNow Predictive Intelligence, Virtual Agent NLU (chatbot)
11.	Infrastructure (Server / Cloud)	Application deployment environment. Local Server Configuration: N/A (SaaS model) Cloud Server Configuration: ServiceNow Now Platform Instance (Multi-Instance SaaS)	ServiceNow Cloud (SaaS), Kubernetes-based MID Server for on-prem integrations

Table-2: Application Characteristics

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Front-end widget rendering and client scripting used inside the Service Portal.	AngularJS, Bootstrap, jQuery
2.	Security Implementations	Access control, authentication and data protection for employee & asset data.	SAML 2.0 SSO, OAuth 2.0, Role-Based ACLs, AES-256 Encryption at Rest, TLS 1.2, OWASP guidelines
3.	Scalable Architecture	Justification of scalability using ServiceNow's multi-instance, workflow-driven design.	ServiceNow Multi-Instance SaaS Architecture, Modular Flow Designer Workflows
4.	Availability	Justification of application availability through cloud redundancy and SLAs.	ServiceNow 99.5% Uptime SLA, Automatic Failover, Distributed Data Centers
5.	Performance	Design considerations for handling concurrent requests and response time.	ServiceNow Platform Caching, Indexed CMDB Queries, CDN for Portal Assets

References

<https://www.servicenow.com/products/product-catalog.html>
<https://developer.servicenow.com/dev.do>
<https://c4model.com/>
<https://www.ibm.com/cloud/architecture>